

Yogendra A

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Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C, HTML, CSS

Frontend & Backend: React.js, Next.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, AsyncIO

AI / ML: PyTorch, TensorFlow, HuggingFace, Whisper, OpenCV, MediaPipe, FAISS, LangChain, scikit-learn

Databases & DevOps: MySQL, MongoDB, Firebase, Docker, Git, Linux, CI/CD, AWS (basic), GCP (basic)

Work Experience

AI/ML Engineering Intern — Audio Intelligence | L&T Technology Services (LTTS) Feb 2026 – Present

- Architected a fully offline **Edge Audio Intelligence Framework** running **14 parallel AI tasks** — Whisper ASR, speaker counting, emotion recognition, language ID across 107 languages, and sound classification across 527 categories; deployed cross-platform on Windows + Linux with zero cloud dependency.
- **Achieved 98% reduction in inference time** (104s → 1.8s) via ThreadPoolExecutor parallelism, model distillation, and pipeline re-architecture; **reduced model footprint by 83%** (6 GB → 1.1 GB) replacing heavyweight transformers with optimised 170 MB alternatives.
- **Cut CPU load by 60%** replacing a TensorFlow deep learning model with an RMS-energy DSP algorithm; patched TorchAudio, SpeechBrain, and HuggingFace source code for stable cross-platform operation.

Projects

Pranayama Trainer — Real-Time Breathing & Posture Analysis [🐙 GitHub](#) Dec 2024 – Mar 2025

- Architected **dual-stream inference pipeline** (webcam + audio, sub-100ms latency) using Python AsyncIO + WebSockets; MediaPipe + OpenCV CV pipeline extracts geometric spinal alignment features in real time.
- Built **React frontend** with AudioWorklet for non-blocking audio capture; persisted session metrics to **Firestore**; replaced TensorFlow model with DSP algorithm achieving **60% CPU reduction**.

VoiceRAG — Voice-Driven RAG Knowledge Assistant [🐙 GitHub](#) Jan 2025 – Mar 2025

- Built end-to-end **RAG pipeline** — document parsing, semantic chunking, FAISS vector search — exposed via **FastAPI** with Streamlit dashboard; integrated **Groq LLM API** for generative answer synthesis over retrieved context; engineered confidence scoring to flag low-relevance outputs.
- **Containerised full pipeline with Docker** for reproducible deployment; LangChain agent architecture with tool-use for dynamic query routing.

Driver Drowsiness Detection & Alert System [🐙 GitHub](#) Jun 2024 – Aug 2024

- Real-time **computer vision system** computing Eye Aspect Ratio (EAR) from 468 MediaPipe facial landmarks; triggers graded hardware alerts via **Arduino (PySerial)** when eyes closed ≥ 3 seconds.
- Cloud event logging to **ThingSpeak IoT API** — blink counts and drowsiness timestamps; deployed on low-resource hardware with sub-50ms detection latency.

Recruitment Management System [🐙 GitHub](#) 2024

- Built a **full-stack web application** for end-to-end recruitment workflow — job posting, candidate applications, status tracking, and HR dashboard — using Python/JavaScript backend + **MySQL relational database**.
- Implemented **role-based access control (RBAC)** for HR managers and candidates; RESTful API design with clean MVC architecture.

Education

PES University, Bengaluru Sep 2022 – May 2026

B.Tech in Computer Science & Engineering | CGPA: 7.05 / 10

Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Operating Systems, Computer Networks

Certifications

- **Goldman Sachs Software Engineering Virtual Experience** — IT security simulation; identified outdated password hashing via Hashcat; proposed algorithm migration and minimum password length policy.
- **Webflow 101 Certification** — Visual web design, component-based layouts, animations, and responsive UI.